Account manager for Disha Kamra

IaC (Infrastructure As a Code)

→ Setup the Infra using the Code.

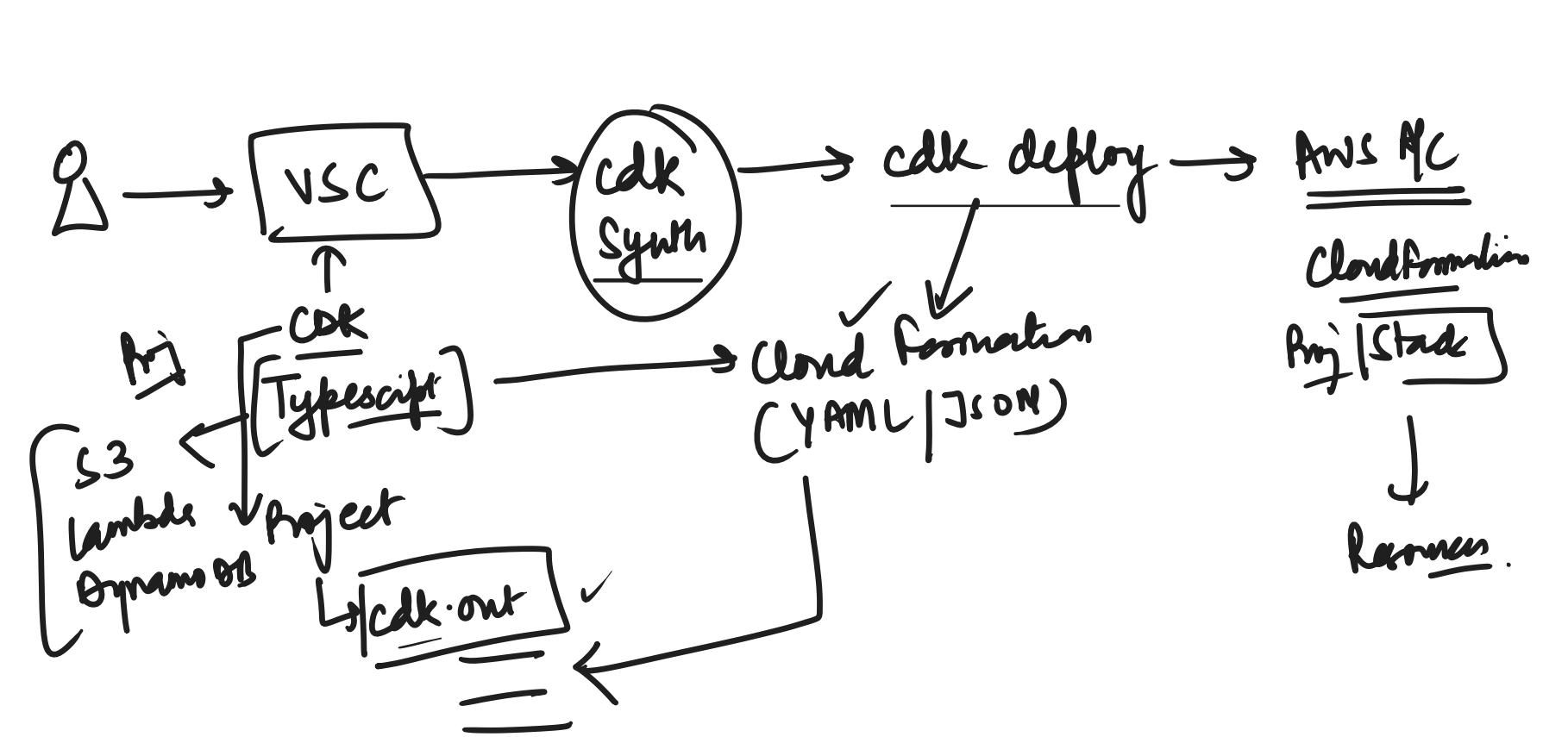
Infra → OnPrem, Cloud*

Cloud ↓ AWS

✓AWS → Compute – EC2, ECS, EKS, lambda
Storage – S3, EBS, EFS, etc.
Networking – VPC, Route Table, SG, NACL, etc.
Databases –

↑
CF
↳ JSON
→ YAML

AWS CDK (Cloud Development Kit) → IaC



L2 – High-level, nicer APIs with sensible defaults.
S3 Bucket → name, versioning, encrypt., Public.
L1 – Direct 1:1 mapping to Cloudformation Resources.
CfnBucket

Ex: Lambda needs write access to S3

⇒ uploadsBucket.grantWrite (uploaderFn); ③ ②

↳ IAM Policy + Role attachment.

Ex: API Gateway calls lambda.

```
const api = new apigateway.LambdaRestApi (this,
'uploadsAPI', {
   handler: uploaderFn,
});
```

① ②

Lab 2 → Constructs

S3 Buckets (L2)
DynamoDB (L2)
Lambda func (L2)
grantWrite to S3
grantWrite Data → DynamoDB
Escape Hatch → enable S3 access logging

What is Context

Context = static Configuration values
↳ cdk.json
↳ cdk.context.json ←
↳ CLI: cdk deploy -c key=value --profile.
↳ Code: this.node.tryGetContext ("key")

lookups

CDK automatically queries AWS to fetch values:
- VPC lookup.
- Hosted Zone lookup
- Load Balancer lookup
- AMI
- AZ

[A/C – Profile – us-east-1]

stack

```
const vpc = ec2.Vpc.fromLookup ( this, 'MyVPC', {
     isDefault: True
});
```

when Query?

[cdk synth] → "cdk.context.json"

```
{
   "vpc-provider:account = 123456789:region= us-east-1": {
        vpcid: "vpc-...",
        az = [—, —]
   }
}
```

Feature flags.

CDK has flags to opt into new behavior safely

cdk doctor:

↳ Enable in cdk.json

TAGs — Cost allocation, Governance, Security, workload (dev/Stage/Prod)

TAG key	Value	Purpose
Env	dev | test | Prod	Stage-Based.
owner	Team-name	Accountability
CostCenter	12345	Billing
Application	PaymentsAPI	Resource grouping

CDK Tagging

* Stack level

```
cdk.Tags.of (this).add ('Env', 'dev');
```

this ↑ current stack

Construct level

```
cdk.Tags.of (myBucket).add ("Dataclass", "Confidential');
```

* App level

```
cdk.Tags.of (app).add ('owner', 'PlatformTeam');
```

ORG & IaC Standards.

Pre-deploy Checks / Guardrails:
- ✓ Block public S3 Buckets
- ✓ Validate naming
- ✓ Ensure encryption-by-default
- ✓ Req TAG mandatory.

Pre-deploy = Aspects

App → CDK Code + Env.
Stack → one CloudFormation / env.
Stage → group of stacks / env
pipeline → Synth, Build, deploy.